

An urban labor market with frictional housing markets: theory and an application to the Paris urban area

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Abstract

We build a tractable model of frictional labor markets and segmented housing markets to study welfare effects of regulations, including spatial misallocation and deviation from competitive pricing of rents. The model is summarized by a labor demand curve depending on rents and wages, a wage curve reflecting labor market tightness and rents, and finally a rent curve reflecting employment. In this economy, the rent gradient in the flexible rent sector is higher than in a purely competitive housing market. This leads to spatial misallocation due to some employees commuting too much and some non-employed living inefficiently close to jobs. In turn, reducing generalized commuting costs reduces the rent gradient in the flexible rent sector and the cost of spatial misallocation of workers. The reduction in market rents is maximal when labor markets are less frictional and housing markets are more frictional, and welfare gains are larger when both are more efficient.

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1. Introduction

In large urban areas, transport infrastructure investments are massive. The monocentric model, which is the workhorse of urban economics after the seminal contributions from Von Thunen (1826) and Alonso (1964); Muth (1969); Mills (1967); Fujita (1989) led to the understanding that, in a *perfectly competitive market*, generalized transportation costs and amenities determine the slope of housing rents in the city. Improvements in transportation efficiency are expected to reduce the rent gradient with a perfect pass-through. Comprehensive reviews of what we know on the impact of infrastructures can be found in Duranton and Puga (2015) and Duranton and Turner (2012).

When applying this standard analysis to most urban areas, however, one needs to take into account the existence of imperfectly competitive housing markets. Indeed, high rents led policymakers to favor social housing and rent controls, resulting in segmented housing markets, possibly quite far away from the competitive paradigm. In addition, the supply of

housing and new constructions is often limited, as is the height of buildings. Both contribute to urban sprawl, with adverse consequences.¹ All in all, in the rental sector, neither rents nor the supply of rental housing units can fully adjust to their competitive equilibrium level. These regulations may have their rationale (Arnott, 1995) but they also segment the housing market and interfere with other policies in a less well understood manner.

In this article, we build a theory of dual housing market and study the properties of its equilibrium when frictions in the labor and housing markets coexist. We also study the role of transportation costs which is different from what is predicted in the competitive paradigm. The model is built to fit a particular example, Paris urban area which is one of the largest in the Western world.² Most of the assumptions made for this city can be used for other areas.

As an illustration of this segmentation, we report in Figure 1 an investigation of the price and rent structure in Paris urban unit. For that, we use a unique original dataset in scraping all rental ads in the urban unit. This and other data sources allow for the analysis of price and rent gradients net of hedonistic characteristics. Figure 1 reports these various statistics, the center of Paris is arbitrarily chosen to be Notre-Dame de Paris. The ratio of the top price/rent in the center to the periphery is near 4:1 for the buyer-seller market. It is 3:1 for new leases in the private rental sector. It is 1.5:1 in the ‘old leases’ sector. It is finally only 1.1:1 in the social housing sector which alone represents more than a quarter of the population in the area. Figure 1(e) reports the fitted rents or transaction prices with 1 as a basis in the periphery. Considering at least two distinct segments of the housing sectors should improve the analysis of any urban policy. Finally, Figure 1(f) also documents the distribution of the social housing sector by distance from the city center. One can observe that it is invariant with respect to distance from the city center and instead rather dispersed across municipalities.

This article therefore seriously takes the existence of a dual housing market featuring the coexistence of a controlled/regulated rent sector with *sub-competitive rents* and of a flexible rent sector with *competitive rents*. By ‘controlled/regulated’ rent sector, we will indeed mostly think of social housing where rents are administratively determined. However, there are also regulated rents for tenants in the private sector, when they have sufficient seniority: the existence of rent controls limiting rent growth *de facto* leads to a divergence between old rents and new (market) rents, thus isolating them from competitive market forces.

In contrast, by ‘flexible rent sector’, we mean the property market (homes and apartments for sales) where prices and transactions are typically free; and new rental transactions, since in most countries and cities, rents adjust to market equilibrium for new leases and are determined in equilibrium.

We develop a flexible framework, which allows for these various interpretations of rent controls and flexible rents. It can serve as a tool to analyze various policies: the additional complexity of the analysis is indeed confined by a set of simplifying assumptions that have to be weighted against the resulting better description of the urban environment as compared to a frictionless housing market.

1 See Glaeser and Kahn (2004) for a documented and vivid criticism of urban sprawl.

2 The Paris agglomeration ranks fourth after NYC, Moscow and Los Angeles. It is the 31st in the world if one includes African, Asian and South American cities. The Paris area consists of 1260 municipalities of the Paris urban area and the 20 districts of Paris ‘*intra muros*’.

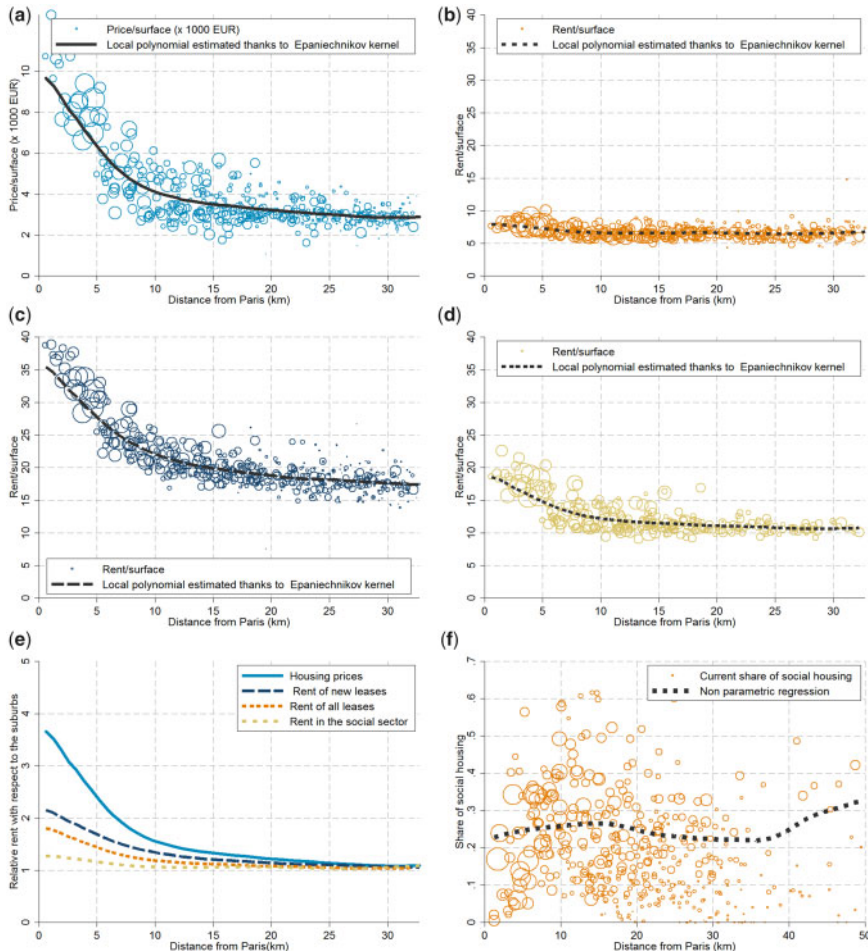


Figure 1. The dual housing sector and its rent gradients: variation of average municipal rents or transaction prices (in EUR per square meter) with respect to the distance from the CBD in the municipalities of Paris Urban Unit (a–e) and share of social housing (f). (a) Transaction price per square meter (real estate), (b) rent per square meter (social housing), (c) rent per square meter (new leases from online ads), (d) rent per square meter (average private rental), (e) fitted rent gradients. Base 1 in the periphery (approximated by the 40–50 km average) in different housing sectors, (f) current share of social housing in the Paris Urban Unit.

Notes: In the calibration, individual transaction data or vacancy lease data are used; the aggregation at the city level in figures is only for presentation convenience. The size of the circle corresponds to the population of the municipality. See [Online Appendix C.3, C.4 and C.5](#) for the details.

Sources: Authors' computations from various data sources: Base d'Informations Economiques Notariales (BIEN) for real estate prices, online ads for new leases, Côte Callon (<http://www.editions-callon.com>) for the average private rental sector and the Répertoire du parc locatif social (RPLS) for the social housing sector and French census for the share of social housing.

The first building block is the dual structure of the housing market. The model features that rent controls produce a feedback effect in the flexible rent sector even though they don't apply there. The effect is complex. It mostly comes from the scarcity of land, but also partly from a feedback effect on wages.

The second building block of the model is a rich labor market where wages reflect the monopsony power of firms and the outside option of workers. This second layer of frictions in the labor market is essential here, because it captures the main effect of housing market segmentation, the spatial misallocation: the employed workers are prevented from accessing some locations close to their jobs since the competitive allocation is partly broken. Our model can, therefore, capture and quantify a misallocation effect of rent controls on commuting distances as illustrated in [Chapelle et al. \(2019\)](#).

In short, we introduce rent regulations in the housing market in the matching approach of the labor market exposed in [Pissarides \(2000\)](#), itself adapted to the urban equilibrium approach of [Fujita \(1989\)](#) as in [Wasmer and Zenou \(2002\)](#). See [Zenou \(2009\)](#) for a review and many extensions. In our model, eliminating both housing market frictions (rent controls) and labor market frictions (matching) leads us back to the Alonso–Muth–Fujita type of urban configurations; eliminating housing market frictions leads back to the benchmark urban search equilibrium; and eliminating only labor frictions leads back to the insightful literature of rent controls (a significant example is [Arnott \(1995\)](#) who demonstrated that rent controls are complement to other market imperfections). In the flexible rent sector, as in [Zenou \(2009\)](#), workers move upon losing their job or getting one. This is an efficiency condition, quite convenient, although not always realistic. Our model improves on this: a large share of workers in the regulated rent sector (and notably those interpreted in the data as under rent controls) do not need to move and thus can stay resulting in an inefficient allocation.³

We then study the properties of the equilibrium in both markets. In this economy, the coexistence of a flexible rent sector and a regulated rent sector distorts price determination. We show that the one-to-one relation between generalized transportation costs and the slope of rents in the city that we should observe in a competitive environment is no longer true.

Frictions in the housing market lead to spatial misallocation, increase in the rent gradients in the urban unit and increase in the marginal effect of transportation costs as well, in a way that we characterize. It is due to some employees commuting too much and some non-employed inefficiently close to jobs. Reducing generalized commuting costs help reduce the rent gradient in the flexible rent sector, raises urban employment and attracts more population, and reduces the cost of spatial misallocation of workers. Wage determination implies some wage–rent feedbacks through original mechanisms. On the one hand, rents in the flexible rent sectors appear relatively inelastic with respect to the average wage but depend on the wage gradient. On the other hand, wages are sensitive to the variation in the maximum rents: land enters the production function of firms; since firms and workers compete for land, the rent distortions are transmitted to wages. We display the relevant curves for three variables (labor market tightness, wages and rents), namely a labor demand curve, a wage-setting curve and a rent curve. These interactions between the labor and the housing markets generate an interesting complementarity between these mechanisms and the level of frictions in the labor and the housing market: the reduction in rents due to better transportation is maximal when labor markets are less

3 We focus on location choices in different areas of the city and in different housing sectors. Intra-area choices are also interesting and could be modelled in a different way, with matching frictions as in [Wheaton \(1990\)](#). We did not pursue this idea here and focus on a dual housing sector where within each sector, the distance to the center is a sufficient statistic.

frictional and housing markets are more frictional, and the welfare gains are the largest when both markets are more efficient.

The article is organized as follows. In Section 2, we provide elements in the French context, followed by a short literature survey to connect our article to the major strands of labor and urban literature. Next, we describe in Section 3 the frictional unemployment model. In Section 4, we derive its main equilibrium properties and derive some comparative statics using realistic parameters for the Urban Unit of Paris. Section 5 concludes.

2. Context and position in the literature

We first document the realism of the key assumptions of the model and therefore describe the housing market context of a large metropolitan area, the Urban Unit of Paris. Most large cities have similar characteristics in terms of dualism of the housing sector and frictions in the labor market, so that the context is not very specific to Paris.

2.1. Context

The housing sector in the Paris area has become increasingly regulated. The share of social housing has increased over the past decades. Its mobility rate, already 4 times lower than in the private rental sector, has decreased.⁴ The allocation into the social housing sector is, if not random, to some extent disconnected from income in France: the income eligibility criteria allows up to 80% of households to be eligible for social housing units ([Cour des Comptes, 2017](#), 25). This, of course, does not mean that richer households have greater chances to access social housing than poorer households, but *ex post*, it turns out that there are indeed households from all income deciles including the top 1 and 2 deciles living in the social housing sector. To a large extent, given the excess demand for social housing in the Paris area and the low mobility rate out of it, access to social housing is largely determined by characteristics left outside our theory and that we will treat as an exogenous random factor. It is also worth noting that the social housing sector is also relatively constantly distributed by distance as illustrated in [Figure 1\(f\)](#). This supports our assumption about the constant distribution of the regulated sector.

Further illustration of the dual character of the housing market in this large agglomeration can be found in Figure C.1 in [Online Appendix](#). We also report in this [Online Appendix](#) (Figure C.2) that Paris has a concentration of lower income and unemployment in the suburbs relative to the center, that explains our focus on one particular equilibrium where unemployed workers are concentrated in the periphery, rather than in the center (see [Glaeser et al., 2008](#), for the US case).

2.2. Literature

On the theoretical side, our research question borrows from different streams of the literature in housing and urban economics. On the one hand, our model belongs to the long tradition of monocentric models that have been developed after the seminal contribution from [Von Thunen \(1826\)](#); [Alonso \(1964\)](#); [Muth \(1969\)](#).

4 Some evidence is provided in [Online Appendix](#), e.g. of Figure C.1(a).

On the other hand, we investigate the impact of partial rent controls as [Fallis and Smith \(1984\)](#) and [Marks \(1984\)](#) who model the housing market as a relatively standard supply and demand system or as [Anas \(1997\)](#) and [Ménard \(2009\)](#) who investigate the impact of rent controls modeling the housing market with search and matching models. There have not been many attempts to introduce rent controls or partial rent controls in a monocentric setting in the urban economics literature. In the most comprehensive analysis of the urban and spatial models (e.g. [Ioannides, 2012](#); [Fujita and Thisse, 2013](#)), the focus is more on public good provision, or on other deviations from competitive product markets such as production or social externalities. We leave these important normative questions aside here and adopt a more descriptive framework of labor and housing market imperfections.

Rent controls are the main source of segmentation of the housing markets. As shown in [Glaeser and Luttmer \(2003\)](#), their misallocation costs are important. They notably show that, when access to the regulated rent sector is random, households may not live in dwellings with the desired characteristics. We build on this key result. The misallocation is 2-fold. The size of dwellings may not be optimal, as in [Glaeser and Luttmer \(2003\)](#); and, as we develop here, the commute time may be too important for some households.

If monocentric models have been widely solicited to investigate the impact of land use regulation as growth control, minimum lot size or maximum Floor Area Ratio (see [Bucovetsky, 1984](#); [Brueckner, 1990](#); [Engle et al., 1992](#); [Bertaud and Brueckner, 2005](#)), rent controls *per se* have remained relatively ignored by this stream of literature. [Heffley \(1998\)](#) discusses this policy and at the same time accounts for its spatial dimension.

[Glaeser and Luttmer \(2003\)](#) define and measure the main inefficiency of the existence of rent controls: the misallocation arises both from the fact that the equilibrium price cannot arise and that some households willing to demand or supply housing are prevented from doing so at the regulated price; but, in addition, that households are misallocated. They estimate that the misallocation in terms of size of dwelling is large, with 25% of households being misallocated in New York.

The literature on rent controls and social housing remains largely empirical with a focus on rich dynamic equilibrium models, such as [Favilukis et al. \(2019\)](#) or [Sieg and Yoon \(2020\)](#) for an application to the city of New York. In the latter, the authors estimate sizeable welfare gains from accessing social housing, around 17%, and very long queues (up to 18 years in Manhattan). [Diamond et al. \(2019\)](#) exploit quasi-experimental variations in assignment of rent controls and document sizeable effects (the supply of controlled rental housing decreases by 15% probably generating an additional increase in unregulated rents).

[Basu and Emerson \(2000, 2003\)](#) model the effects of a cap on rent increases for existing leases together with an asymmetry of information on the likely duration of a stay by tenants, leading to additional inefficiency in the allocation of space. In particular, an additional adverse consequence is the prevalence of unfair evictions of tenants by landlords.

The spillover between rent controls and housing prices present in our model is an important mechanism that has been documented in the empirical literature: [Autor et al. \(2014\)](#) find that the deregulation of rent controls in the Boston area led to massive capital gains in real estate.⁵ See also [Asquith \(2019\)](#) who investigates the consequences of the cap on rent increases in San Francisco.

5 In their article, rent control reduces rents in the uncontrolled sector, while in our article, the spillover is positive for two reasons that could be identified empirically in future work: scarcity of land implies that more social

Interaction between housing markets and labor markets has generally been studied without introducing bid rents. When they include a proper urban rent structure, there are frictionless housing markets.⁶ We attempt to fill this gap providing a general model of labor and housing frictions with equilibrium rents.

Our open city model is closely related to the literature documenting the impact of transport infrastructures on city growth and urban sprawl. If the impact of transport infrastructures (e.g. [Gobillon et al., 2007](#)) on employment has been studied very accurately, more is to be understood on the general equilibrium interactions between transport infrastructures and housing market regulation on urban area growth, segregation, unemployment or productivity and their quantitative properties. For example, in the USA, several contributions documented the impact of transport infrastructures on suburbanization as in [Baum-Snow \(2007\)](#) or employment and productivity growth ([Duranton and Turner, 2012](#); [Chatman and Noland, 2014](#)).

We are also related to empirical papers studying the impact of transport infrastructures on city structure, surveyed comprehensively by [Redding and Turner \(2014\)](#). [Kahn \(2007\)](#) documents that neighborhoods with better connection to mass transit experience a faster gentrification process. Our model also has implications for the impact of transport infrastructures on commuting patterns, land prices and welfare. These aspects have recently been documented in [Severen \(2018\)](#). He estimates the empirical effect of mass rail transit on commuting flows, housing prices and welfare in a very different setting: the city of Los Angeles characterized by a very low density. Our model provides an analysis of the connection between transport infrastructures and land/housing price as in [Severen \(2018\)](#) and [Gibbons and Machin \(2005\)](#), with a focus on the rent gradient as [Ahlfeldt and Wendland \(2011\)](#) and [Ahlfeldt \(2011\)](#). As our model is monocentric, we leave aside the fact that firms might relocate after a change in transport infrastructures as this phenomenon remains relatively limited as documented in [Mayer and Trevien \(2017\)](#); [Garcia-López et al. \(2017a, 2017b\)](#) for Paris.

Regarding data collection, we follow [Combes et al. \(2018\)](#) and use transaction prices collected by the French notaries. We also follow emerging literature scraping data from realtors' websites. For example, [Mense et al. \(2017\)](#) use scraped rental data to assess the impact of rent controls in Germany, while [Horn and Merante \(2017\)](#) use a dataset gathered by a private provider to estimate the impact of Airbnb on the rental market. Finally, [Brühlhart et al. \(2017\)](#) use this kind of data privately provided to estimate the tax incidence of property tax in Switzerland. Our results based on these diverse sources of data tend to confirm the discrepancy between the gradient measured with housing price or with rental data documented in [Halket et al. \(2015\)](#) for London.

3. A frictional labor market in a simplified model of housing

This section introduces the model and describes its equilibrium. Time and space are continuous and unidimensional. The city is defined by a Central Business District (CBD) and a city border indexed by a distance to the center and normalized to 1.

housing reduces the supply of private housing. Further, more rent controls for existing leases lead to higher rents of new leases. A large fraction of the capital gain of deregulation came from never-controlled units that gentrified. Our article does not focus on spillover effects from the composition but rather on pecuniary externalities arising from the scarcity of space.

6 For example, [Wasmer and Zenou \(2002\)](#), an exception being [Wasmer and Zenou \(2006\)](#).

There are two types of agents, households and firms. Households are also workers and may either be employed or unemployed. Workers are located in one of the two types of housing sectors: the first one has rents determined by the market: hereafter it is called the *flexible rent (FR) sector*. The second one has rents that are controlled and is called the *regulated rent (RR) sector*. Both housing sectors will be populated in equilibrium by employed and unemployed workers. Hence, *ex post*, there are four categories of agents depending on their employment status (employed or unemployed) and the status of the housing sector (in the RR sector or FR sector). There is also the underlying dimension of space heterogeneity, the distance from the center, which will affect the workers' income and their utility. We, therefore, abstract away from more complex *ex ante* workers heterogeneity issues analyzed in Diamond (2016) in order to focus on the insights from this simpler model with no heterogeneity *ex ante*. All firms are located in the CBD and hire workers from the four groups.

Households change employment status randomly, according to a matching-and-separation process described by Poisson shocks. We have explored different wage-setting mechanisms but our preferred one is a bargaining model combining endogeneity of wages and a wage-rent transmission mechanism. Workers also face reallocation Poisson shocks leading them to change housing sectors or to move to other parts of the city. Finally, they quit the city exogenously and are replaced by inflows of new households ensuring stationarity. These new households move either in the RR sector to replace those who left (this is a queuing process) or in the FR sector. In all cases, the newcomers start unemployed and get a job only from the city.

We can now introduce the main notations.

3.1. Basic notations

We follow here Wasmer and Zenou (2002) and add up to the standard monocentric model, a matching process in the labor market. By probability per unit of time, we hereafter refer to Poisson parameters. The city has a center called Central Business District (CBD). Households live at a distance dL , where total population in the city is L , and d is an index of distance from the CBD varying between 0 and 1. With fixed population, we normalize population to $L = 1$. With endogenous population (see Section A.4 in Online Appendix), a larger population will result into larger commuting distance for households living at a distance $d > 0$, linearly increasing with respect to L .

3.1.1. Discounting

Agents discount time at a rate r_0 . In addition, workers leave the city—or die—at Poisson rate Ξ . It indeed acts as a death rate as in perpetual youth models à la Blanchard (1985). We normalize the value of exiting the urban area to zero. Hence, the total discount rate r is the sum of the *pure* discount rate r_0 and of the inter-city reallocation shock Ξ : we hereafter use notation $r = \Xi + r_0$ for this total discount factor.

3.1.2. Labor market: matching, separation and wages

An endogenous number of unemployed u looks for jobs and are matched with an endogenous number of vacancies V through a matching technology function $M(u, V)$ that has constant returns to scale. Let labor market tightness $\theta = V/u$ be the ratio of each side of

the market and $f(\theta)$ be the hiring probability (per unit of time). The total instantaneous number of matches is a Poisson parameter. It depends on the function M of the total number of unemployed u and vacancies V . Formally,

$$f(\theta) = \frac{M(u, V)}{u} = M_0 \theta^{1-\eta}.$$

where the last term is, to fix ideas, the specific case of a matching function that is Cobb–Douglas with a scale parameter M_0 and an elasticity of unemployment η . Conversely, firms have a per period probability to meet a worker. This probability per unit of time is the parameter of a Poisson process denoted by $q(\theta)$ with $q' \leq 0$ and defined as

$$q(\theta) = \frac{M(u, V)}{V} = M_0 \theta^{-\eta}.$$

with again a specific Cobb–Douglas case.

Jobs are characterized by exogenous labor productivity denoted by y . Firms willing to recruit pay a search cost per unit of time γ . Jobs end up at a Poisson process of intensity s and by the departure from the city at rate Ξ . The income of unemployed is z . We will denote by $w^f(d)$ the wage of agents living in the FR sector, and $w^r(d)$ the wage of agents in the RR sector. Interestingly, the two wage profiles differ quite a lot *ex post*, as we will discuss later on.⁷

3.2. Reallocation shocks within the city

Workers in the FR sector may eventually be hit by a reallocation shock and access the RR sector. This features quite well, at least in France, the accession to social housing (people are typically on waiting lists for 2–5 years) and as already argued, 80% of the population is eligible.⁸ It also features well the effect of rent controls in the FR sector: households start at the competitive rent in a new lease and then get protected from market pressures because the rent cannot be re-adjusted within the current lease.

Specifically, we assume that with Poisson intensity δ_0 , the worker in the FR sector leaves the dwelling and is reallocated across housing sectors with probability p^r in the RR sector and probability $p^f = 1 - p^r$ in the FR sector, but remains in the same area, or technically, at the same distance (e.g. in the city of Paris, the household may move from close suburb Clichy to close suburb Malakoff but at the same distance from the center). We assume that workers in the RR sector are affected by the same shocks at the same rate δ_0 . In [Online Appendix](#), we let the intensity of arrival to differ to be slightly more general. These assumptions lead directly to the fact that the relative size of the RR sector is proportional to p^r , and actually equal to p^r when we assume the same Poisson arrival rate δ_0 .

⁷ There is a small dependence of wages to distance but quite a marginal one quantitatively as discussed later on with realistic parameters. Quantitatively, we find that the slope of wages with respect to distance is small, and much less steep than the cost of commute. In other words, most of the cost of commute is compensated by lower rents, and very little by wages. We can even decompose this relative effect. With our calibration, wages explain 3.8% of the variation of rents within the city while transport costs explain as much as 86%. The residual part is accounted for by the expected gains from transitions, which are themselves a mix of the two effects discussed above in the next location, so more or less in the same proportions.

⁸ The actual proportion of eligible tenants is even higher, in fact, since homeowners are richer and typically do not apply for housing, thus reducing the denominator in the eligibility rate.

for workers in each sector. Note that in all proofs in [Online Appendix](#), we also consider the case of a reallocation of workers in the city introducing a supplementary reallocation shock with another Poisson intensity δ_1 . Since this shock is not essential here, we present the model with $\delta_1 = 0$.

3.3. Competition for space

In the FR sector, households of different employment statuses compete given their bids. Households bid for rents at a higher level in the FR sector when rents in the RR sector are themselves lower because they expect to reduce the rental cost later on when they access the RR sector. This leads to an equilibrium in which, close to the CBD, winners are the employed workers; in areas distant to jobs, unemployed have instead a higher marginal propensity to pay.⁹ Bidders are *ex post* indifferent across all locations given their bid, as is standard in urban economics. In the RR sector, agents are assigned to a location and pay an exogenous rent. A preview of the equilibrium can be seen in [Figure 2](#).

Overall the model highlights a rent–wage feedback: wages respond to rents in complex ways and illustrate a complementarity.¹⁰

3.4. Value functions (expected present discounted values of income streams)

Denote by $W_U^j(d)$ and $W_E^j(d)$, the present discounted value (PDV) of living at d in each state (unemployment and employment, respectively) and each housing sector, where $j = r, f$ stands for the RR and FR sectors, respectively.

One can define their expectations after knowing the sector type but before knowing the precise location as $\mathbb{E}_{d,j} W_E^j$ or $\mathbb{E}_{d,j} W_U^j$ for $j = r, f$. The operator $\mathbb{E}_d$ is the expected value after reallocation in the urban area given the shock and the change in the employment status.

Under linearity in d that we will verify *ex post*, these value functions are linear in commute cost and thus in distance. The value functions are then simply a function of the average distance given their employment status and housing sector. In the FR sector, individuals are sorted according to their employment status. In the data, not all labor transitions may lead to relocation, however, [Delance and Vignolles \(2017\)](#) find a robust and positive correlation between unemployment shocks and the probability to change of housing unit, controlling for many socio-demographic and local characteristics.

We define and study an urban equilibrium ‘EU’ where the employed workers in the FR sector bid away the unemployed. The average distance to the center of the employed and unemployed workers in the FR sectors are denoted as $\bar{d}_e = (1 - u)/2$ and $\bar{d}_u = 1 - u/2$.

9 We will mostly study the equilibrium where employed workers living in the FR sector are all uniformly closer to the center than the unemployed in the FR sector. This is not the only equilibrium and that another equilibrium may arise, where the unemployed in the FR sector prefer to reside closer to the center.

10 In a previous version of the article, we used a different mechanism importing the Rogerson-type lotteries that are widely used in the quantitative macroeconomic literature; see e.g. [Rogerson \(1984, 1988\)](#); [Rogerson and Wright \(1988\)](#), [Greenwood and Huffman \(1987, 1988\)](#), [Hansen \(1985\)](#); [Hansen and Sargent \(1988\)](#) or [Shell and Wright \(1993\)](#). Agents in this literature typically commit to a labor supply before knowing the realization of a main random shock affecting labor demand. Hence, prices are *ex ante* competitive but *ex post* are not; yet agents fulfill to their *ex ante* commitments and subsequently do not deviate, at least during a short run, before some new sequence starts. Although more elegant, the assumption may be seen as less relevant in housing markets. Yet, results with the assumption of the present text and under the lottery assumption of the previous text are very similar. Previous results are available upon request.

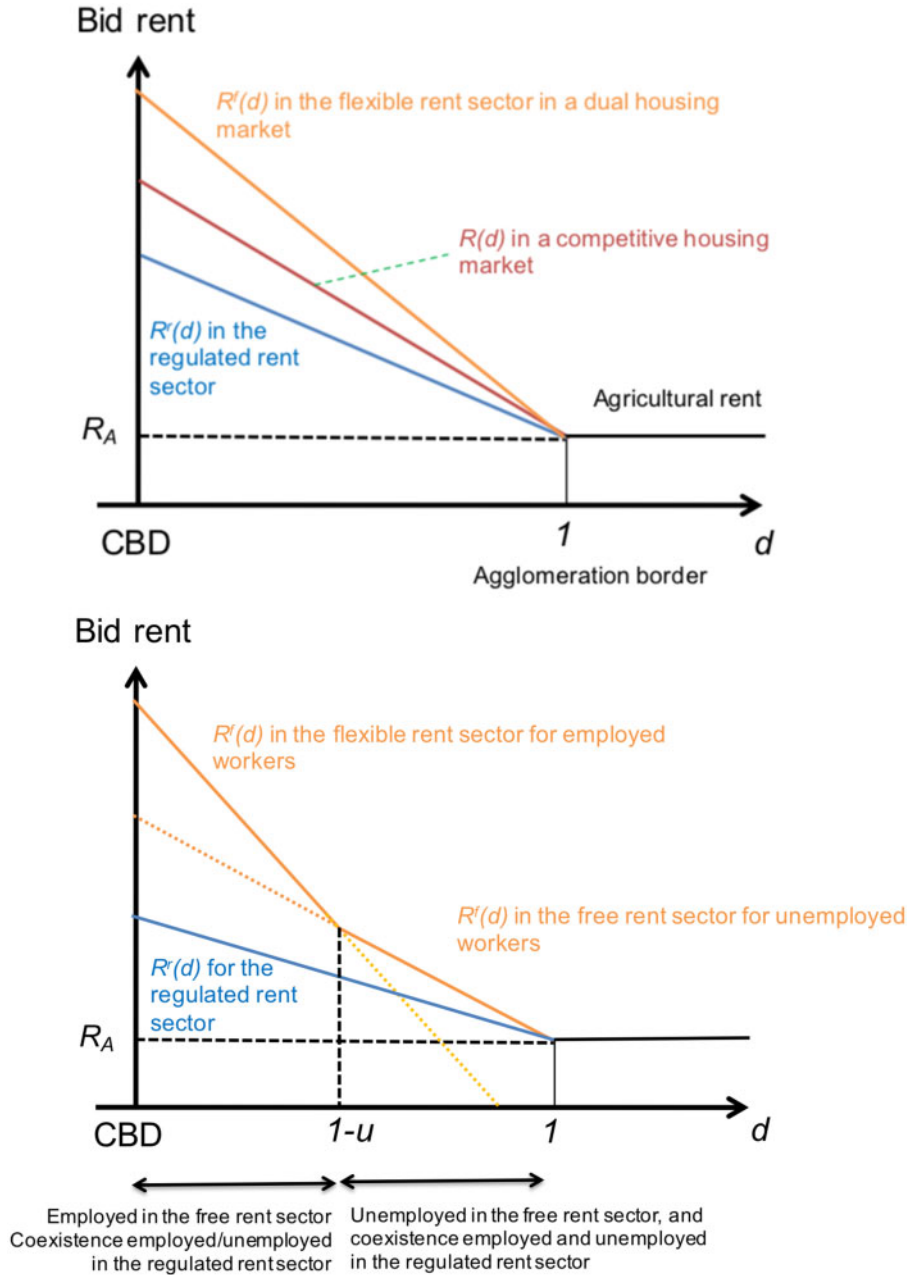


Figure 2. Rent equilibrium in the urban model augmented with housing market frictions. Top: competitive labor market, with absolute values of the slope in decreasing order

$\tau_e + \frac{\delta_0 p'}{r + \delta_0 p'} \left[\tau_e + \frac{\partial R'(d)}{\partial d} \right] > \tau_e > \frac{\partial R'(d)}{\partial d} \Big|_{w^r(d)}$; Bottom: frictional labor market, with absolute values of the slope in decreasing order $\tau_e + \frac{\delta_0 p'}{r + \delta_0 p'} \left[\tau_E + \frac{\partial R'(d)}{\partial d} \right] > \tau_u + \frac{\delta_0 p'}{r + \delta_0 p'} \left[\tau_U + \frac{\partial R'(d)}{\partial d} \right] > \frac{\partial R'(d)}{\partial d} \Big|_{w^r(d)}$.

3.5. Determination of ex post asset values

We can now describe the asset values of workers.

Generically, we define the conditional expectation operator $\mathbb{E}_{sec} W_K(d_r, d_f)$ over asset values W_K^j for $j = r, f$ and $K = U, E$ and as a function of future expected distances in each housing sector (hence the subscript *sec*), respectively, d_r and d_f , as

$$\mathbb{E}_{sec} W_K(d_r, d_f) = p^r W_K^r(d_r) + p^f W_K^f(d_f) \quad (1)$$

this quantity defines the expected value across sectors for a given distance in each of the housing sectors.

This notation $\mathbb{E}_{sec} W_K$ will be very useful in other key statistics of the model and we use them consistently later on. The unemployed and the employed pay different commuting costs τ_u and τ_e , respectively, pay a flexible (market) rent R^f or a regulated rent R^r depending on the sector $j = f, r$,¹¹ receive an income z (for the unemployed) and $w^j(d)$ (for the employed). Wages potentially depend on distance d and sector $j = r, f$.

Agents in the FR sector optimally relocate to the urban area during employment to unemployment or unemployment to employment transitions. We therefore have for workers in the FR sector:

$$\begin{aligned} rW_U^f(d) &= z - \tau_u d - R^f(d) + f(\theta) [\mathbb{E}_{d|f} W_E^f - W_U^f(d)] \\ &\quad + \delta_0 [\mathbb{E}_{sec} W_U(d, \bar{d}_u) - W_U^f(d)] \end{aligned} \quad (2)$$

$$\begin{aligned} rW_E^f(d) &= w^f(d) - \tau_e d - R^f(d) + s [\mathbb{E}_{d|f} W_U^f - W_E^f(d)] \\ &\quad + \delta_0 [\mathbb{E}_{sec} W_E(d, \bar{d}_e) - W_E^f(d)] \end{aligned} \quad (3)$$

Equation (2) states that the equity value of being unemployed $r_0 W_U^f$ is the stream of the utility including unemployment benefits, costs of commute at a distance d and the rent paid at that distance, workers make a capital gain if they get a job with Poisson intensity $f(\theta)$, they are reallocated across sectors at rate δ_0 and leave the city and obtain a value zero at rate Ξ which therefore adds up to r_0 in the left-hand side. A similar interpretation holds for the second equation: the equity value $r_0 W_E^f$ takes into account the wage, netted out of commuting costs and rents, the reallocation across housing sectors and the loss of a job in case of a separation shock s and the reset value of the utility upon a shock Ξ .

If agents are in the RR sector, they bargain over their wage, but they stay where they were assigned to in case of disagreement on wages, and thus do not move. This features well the lack of mobility of people in the social sector. They are also immobile across employment to unemployment transitions or unemployment to employment transitions. We therefore have, with similar interpretations, the Bellman equations in that sector:

$$\begin{aligned} rW_U^r(d) &= z - \tau_u d - R^r(d) + f(\theta) [W_E^r(d) - W_U^r(d)] \\ &\quad + \delta_0 [\mathbb{E}_{sec} W_U(d, \bar{d}_u) - W_U^r(d)] \end{aligned} \quad (4)$$

11 Following the literature and to ease the comparison between the sub-segments of the housing market, we transform prices of properties into their annuity value (their ‘imputed rent’), to compare the actual price of land service across housing sectors.

$$rW_E^r(d) = w^r(d) - \tau_e d - R^r(d) + s[W_U^r(d) - W_E^r(d)] + \delta_0[\mathbb{E}_{sec} W_E(d, \bar{d}_e) - W_E^r(d)] \quad (5)$$

4. Description of the equilibrium allocation in space and in the labor market

The equilibrium determination involves several steps. The first one is the equilibrium allocation of workers of different types in the city. This step involves the demand for rents, as a function of the value of being in the city and determines the boundary of each sector in the city. The second step is to derive the equilibrium in the labor market, from the firms' entry. This determines, for a given set of wages and rents, the equilibrium labor demand curve. The third step is to derive a wage curve, that is, a link between wages and other equilibrium variables including labor market tightness and rents.

4.1. Allocation of workers within the urban area and rents

The allocation in space follows a standard bid rent model. Agents willing to live in place indexed by distance d to the CBD issue a bid rent to access the FR sector. Bid rents are conditional on the employment status.

To express bid rents, the agent calculates the value of being in a position d in the urban area. In the absence of mobility cost in the FR sector, the bid rent of each group, employed and unemployed, insures that workers of a given employment status are indifferent between different locations. Any utility differential would indeed be eliminated by the corresponding increase or decrease of the bid rents (see [Wasmer and Zenou, 2002](#), for this property in the standard urban frictional unemployment context).

Denote by $\bar{W}_E^f = \mathbb{E}_{d,f} W_E^f(d) = W_E^f(d)$ and $\bar{W}_U^f = \mathbb{E}_{d,f} W_U^f(d) = W_U^f(d)$, the common asset values of living in the city in the FR sector; the bid rents of the bidders are therefore given by $\Psi_U(d)$ and $\Psi_E(d)$ as:

$$\Psi_U(d) = z - \tau_u d + f(\theta)[\bar{W}_E^f - \bar{W}_U^f] + \delta_0[\mathbb{E}_{sec} W_U(d, \bar{d}_u) - \bar{W}_U^f] - r\bar{W}_U^f \quad (6)$$

$$\Psi_E(d) = w^f(d) - \tau_e d - s[\bar{W}_E^f - \bar{W}_U^f] + \delta_0[\mathbb{E}_{sec} W_E(d, \bar{d}_e) - \bar{W}_E^f] - r\bar{W}_E^f \quad (7)$$

A first insight is that the classical bid rent slopes are determined by τ_e and τ_u to which a part must be added, related to the spatial gradients in utility in the RR sector, by rents in the FR sectors and by wage gradients in both sectors. These additional components all come from a very important mechanism: the absence of a rent adjustment in the RR sector leads to utility differences across areas of the city and will lead the workers luckily assigned to center to benefit from a much higher utility in the center. We denote by R_A the exogenous (agricultural) rent at the periphery of the city, which will fix the value of the endogenous rent profile at distance $d = 1$ (city fringe).

The second property is that this mechanism is anticipated by workers. This implies that the lower the rent in the RR sector and the higher its share, the higher the rent in the FR sector. This will push up rents in the free sector, with a larger impact near the center of the city.

Proposition 1 (slopes of bid rents and transportation costs in partial equilibrium):

1. In a competitive labor market with full employment and in a competitive housing market with no RR sector ($p^r = 0$), the wage is independent of distance to jobs and >the (negative of) the slope of the bid rent $-\frac{\partial \Psi_E(d)}{\partial d}$ is τ_e .

2. In a competitive labor market but with a positive size of the RR sector ($p^r > 0$), wages are still constant in the city, and the (absolute value) of the slope of the bid rent is

$$-\frac{\partial \Psi_E(d)}{\partial d} = \tau_e + \frac{\delta_0 p^r}{r + \delta_0 p^f} \left[\tau_e + \frac{\partial R^r(d)}{\partial d} \right]$$

3. The result generalizes easily to frictional labor markets. Holding wages constant in partial equilibrium, the bid rents are given by, respectively, for unemployed and employed workers:

$$\begin{aligned} -\frac{\partial \Psi_U(d)}{\partial d} \Big|_{w^r(d)} &= \tau_u + \frac{\delta_0 p^r}{r + \delta_0 p^f} \left[\tau_u + \frac{\partial R^r(d)}{\partial d} \right] \\ -\frac{\partial \Psi_E(d)}{\partial d} \Big|_{w^r(d)} &= \tau_e + \frac{\delta_0 p^r}{r + \delta_0 p^f} \left[\tau_E + \frac{\partial R^r(d)}{\partial d} \right] \end{aligned}$$

with $\tau_U = \tau_u(1 - \omega_f) + \tau_e \omega_f$ and $\tau_E = \tau_e(1 - \omega_s) + \tau_u \omega_s$ are convex combinations of commuting costs with weights $\omega_f(\theta) = \frac{f}{r + \delta_0 p^f + f + s}$ and $\omega_s(\theta) = \frac{s}{r + \delta_0 p^f + f + s}$; they both converge to a common value τ when commuting costs are identical for employed and unemployed workers, $\tau_u = \tau_e = \tau$.

4. The rent gradients are monotonically decreasing with commuting costs τ_e and τ_u .

5. The ex post condition for having the employed bidding away the unemployed in the FR sector is that the slope of the bid rent is higher for the employed and in each case, lead to an easy solution that collapses to $\tau_e > \tau_u$ in the specific case $p^r = 0$.

The complete calculations are in [Online Appendix B.4](#). In the top panel of [Figure 2](#), we represent the main mechanisms in a competitive labor market. The slope of the rent function in the RR sector is by assumption a policy parameter. It shifts away the bid rent in the FR sector. This competitive labor market case is obtained from considering $\tau_u = \tau_e$ and letting the hiring probability ($f(\theta)$) go to infinity, that is when both value functions W_U^f and W_E^f converge toward each other, the same being true of W_U^r and W_E^r . Equations describing it are also reported in [Online Appendix B.3](#).

The intuition of (1) above is straightforward: a reduction in transport costs reduces the propensity to pay for being marginally closer to the center as in the standard urban model. As regards to (2), the bid rent in the FR sector is steeper when the term in brackets is positive. This happens when the rent of the RR sector is less steep than the commute cost, which is quite a realistic assumption if the regulated sector has rather flat rents in distance.¹² It is also important to note that *the marginal impact of a decline in transportation costs is greater, the larger housing frictions*. As regards to (3), the possibility of transitions

12 This effect comes from $\delta_0 > 0$ and would disappear if $\delta_0 = 0$. This means that if workers expect to remain in a given distance d in the RR sector (e.g. because they are offered a social housing in that location or because rent controls prevent rents to further increase in the existing lease) then they are ready to pay a higher rent as of today to stay here and capitalize on lower rents in the same place in the future.

between labor market states is reflected in the values of transportation costs: τ_U reflects the expected value of transport costs and would converge to τ_u if the job-finding rate f was zero or conversely to τ_e if the job-finding (Poisson) rate was infinite; the same interpretation holds for τ_E . Overall the effect of frictions in the labor market leads rent profiles to adjust taking into account the future transitions in the labor market, as in Wasmer and Zenou (2002).

In the bottom chart, we depict the equilibrium with labor market frictions. As it can be inferred from the graph, in the FR sector, the employed fully bid out the unemployed from the center¹³ if the rent gradient is higher for them than for the unemployed, which is the case when the commuting costs of the employed are large enough relative to the commuting costs of the unemployed:

$$\tau_e \gg \tau_u \quad (8)$$

We will restrict our analysis to this realistic case only.¹⁴ Intuitively, and as illustrated in Point 5 of Proposition 1, Condition (8) states that for the employed to want to bid out the unemployed, their willingness to pay to be closer to the center by one marginal unit τ_e has to exceed that of the unemployed. Note that the slopes of bid rents also depend on gradients of utilities and wages simultaneously determined in equilibrium, which is why $\tau_e > \tau_u$ is not a sufficient condition for the equilibrium.

Under Condition (8), the highest rent in the center is given by $R^f(0) = \gamma_E(0)$; in the ‘suburbs’ at $d^{sub} = 1 - u$ where the two bid rents are equal¹⁵; in the city border, at $d = 1$, one has $R_A = \Psi_U$.

As a final remark, note that on both charts, we have assumed that the bid rent of the FR sector was equal to R_A , a standard closing condition, and for convenience, that the rent in the RR sector $R^r(1)$ in $d = 1$ was also equal to R_A . This does not need to be the case, and in our comparative static exercises, we will let $R^r(1)$ vary as the outcome of policy without constraining it to be equal to R_A .

4.2. Unemployment rates in different parts of the city

We normalized the urban population L to 1 in the closed city (fixed population) equilibrium and will endogeneize L in the open city equilibrium later on. In the RR sector, some households may be either employed or unemployed, depending on their past trajectories.

13 EU-type equilibrium with the employed in the center as opposed to UE-type equilibrium with the unemployed in the center. Note, however, that the fact that there are two urban allocations does not imply multiplicity, even though this is a theoretical possibility. The strategy of the proof is to study each urban configuration, find the condition on endogenous variables and in particular on θ to have that particular urban configuration, and show that EU and UE are mutually exclusive. See Wasmer and Zenou (2002) for such a strategy in the specific case of a frictionless urban market. In our more general context, the property remains true because the urban configurations only depend on the shape of the FR sector, which itself has the same properties as in Wasmer and Zenou (2002). As regards to uniqueness of the equilibrium once the urban configuration is determined, this is relatively standard and fully addressed in the Online Appendix.

14 If in US cities, the concentration of poverty and unemployment is rather in the center, as detailed in Glaeser et al. (2008), the European cities are rather different with instead wealth and employment in the center and sub-urban concentrations of unemployment, as discussed in Brueckner et al. (1999). For this reason, we do not study here the other equilibrium where the unemployed in the FR sector would bid away the employed from the center as it does not capture well the type of European urban area the model targets but this can easily be done in changing the above definitions of $\bar{d}_e$ and $\bar{d}_u$ into, respectively, $(u + 1)/2$ and $u/2$.

15 Formally, the suburbs start at $d^{sub} = 1 - u^f$ where u^f is the rate of unemployment of the workers in the free sector; however, $u^f = u^r = u$ and the unemployment u^r in the regulated sector is spread in the city.

Interestingly, this dual housing sector structure allows for a non-degenerate distribution of the employed and the unemployed in the urban area. Under the maintained assumption of an EU-type equilibrium (where EU means that the employed are in the center and the unemployed in the periphery in the FR sector, see details in [Online Appendix B.7](#)), as in [Equation \(8\)](#), the inner urban area has a high level of employment and a low level of unemployment u_{inner} , while the periphery has lower employment and a higher level of unemployment $u_{\text{periphery}}$. The values of unemployment rates in the center and the periphery are reported in the last part of [Online Appendix B.1](#).

Aggregate unemployment itself is obtained from the steady-state relationship between inflows into unemployment and outflows, namely, recalling that there is an inter-city reallocation shock Ξ leading to leaving one's job in the city:

$$f(\theta)uL = L(s + \Xi)(1 - u) \quad (9)$$

Replacing, in this equation, tightness by the ratio V/u , it is easy to obtain the so-called Beveridge curve, a negative relation between vacancies and unemployment.

4.3. Computing the welfare costs of rent controls

As illustrated in the previous section, the introduction of the RR sector spreads the unemployed workers throughout the city. This will thus generate a misallocation of workers when compared with the standard competitive framework. Indeed, on average, employed workers commute from the periphery while if they were in the FR sector, they would be closer to the center. Reciprocally, some unemployed workers commute from the center, thus displacing the employed workers who pay higher commuting costs. The total efficiency loss can be measured as the difference between total commuting costs of individuals in the RR sector relative to the counterfactual of them being in the FR sector:

$$\begin{aligned} \mathbb{L} &= p^r \{ [u\tau_u 1/2 + (1 - u)\tau_e 1/2] - [u\tau_u \bar{d}_u + (1 - u)\tau_e \bar{d}_e] \} \\ &= p^r \frac{u(1 - u)}{2} (\tau_e - \tau_u) \end{aligned}$$

using the definitions of $\bar{d}_u, \bar{d}_e$ for the average distance of unemployed and unemployed in [Online Appendix](#) Equations (B.1) and (B.2).¹⁶

A second effect is the one apparent in Proposition 1: the larger the RR sector, the higher the general level of rents, which adds up to the costs of firms and reduces employment. On the contrary, by reducing average rents for workers, it may lead to a more attractive city when the population is endogenous as discussed in Section A.4 in [Online Appendix](#). A companion paper ([Chapelle et al., 2019](#)) explores this issue in detail.

4.4. Free entry of firms and labor demand

At the recruitment stage, firms form expectations on the type of workers they will face and their residency and know that these dimensions impact on their profit through the outcome of the wage determination process, as explained below. Formally, firms meet job

16 Interestingly, the welfare cost due to misallocation using our parameter values applied to Paris urban area is approximately 2% of the total commuting costs paid by workers in the city, of the order of magnitude of the gains expected from the large project infrastructure in Paris (Grand Paris Express).

seekers living in both sectors, with probabilities p^r and p^f . In each housing sector, they find workers at a distance depending on that sector: workers in the RR sector are uniformly distributed in the urban area and workers in the FR sector are uniformly located between the beginning of the ‘suburb’ at $d = d^{sub}$ and the urban area border $d = 1$. We denote by J_V the value of a vacancy and by $J^j(d)$, $j = r, f$ the value of a job with a worker in housing sector j at a distance d , and $\mathbb{E}_{sec}J(d)$ and $\mathbb{E}_{sec}J(\tilde{d})$ the expectation of meeting a worker at a distance d or at an average distance $\tilde{d}$.

Firms also pay an office space per employee. This cost is proportional to the rent paid by tenants in the center, that is, $R^f(0)$. The proportionality is actually the ratio of the office space per employee divided by the dwelling space per active members of households. We think of this ratio, denoted by a , as a technological parameter, fixed in the short run, that is, firms cannot affect it. It is nevertheless possible to introduce land use by firms as another input to enrich the model. We discuss this in Section A.2 in Appendix.

We assume here that the firm pays the full rent in the vacancy stage, to optimally avoid adjustment costs during the recruitment process.¹⁷ The value of a vacancy is then given by:

$$rJ_V = -\gamma - aR^f(0) + q(\theta)\mathbb{E}_{sec}J(\tilde{d}) \quad (10)$$

A free-entry equilibrium implies that $J_V = 0$. Using [Online Appendix](#) Equation (B.11) for the derivation of the asset value of a filled position and the definition of the expected value of a job across sectors in Equation (B.12), the value of a job linked to the labor market tightness:

$$\mathbb{E}_{sec}J(\tilde{d}) = \frac{\gamma + aR^f(0)}{q(\theta)} = \left(\frac{\gamma + aR^f(0)}{M_0} \right) \theta^\eta \quad (11)$$

$$J^j(d) = \frac{y - aR^f(0) - w^j(d) + \delta_0 \mathbb{E}_{sec}J(d)}{r + s + \delta_0} \quad (12)$$

for $j = r, f$. [Equation \(11\)](#) states that the lower the value of search costs, the lower the office cost for firms and the higher the expected value of jobs, the higher the entry of firms and therefore the higher the equilibrium level of labor market tightness. The second equality in that equation is obtained with the specific functional form for a Cobb–Douglas matching function. [Equation \(12\)](#) states that the value of a job is equal to the flow of income, that is productivity net of wages and office costs, takes into account future reallocation shocks and separation shocks and is discounted by the sum of the discount factor and of the various separation shocks.

4.5. Wage determination

We can now complete Proposition 1 with a discussion of wages. It is useful to recall that with frictions in the labor market, wages are not determined competitively. In a competitive market, productivity, the reservation wage of workers and their wage are all equal. In a frictional market, one will have a wedge between the reservation wage and productivity. Bargaining over wages implies the wage to fall between the reservation wage and

17 Since the spirit of the matching model is to feature the marginal decision of a large firm, the PDV of a vacancy is also the PDV of a marginal vacancy of a large firm that may not be able to adjust its office space during the time interval of hiring without the large adjustment costs referred to in the text. See [Online Appendix](#) subsection B.2 for a full discussion.

productivity. We detail this wage determination now and explore whether wages also reflect the rent and the commuting costs, and how. Indeed, part of these values is private information to the worker, and might not be credibly revealed by tenants. Empirically, commuting costs affect wages (see e.g. [Guglielminetti et al., 2016](#)), and thus we assume they enter in the bargaining equation: this is why, so far, we let wages be potentially a function of distance $w^j(d)$.

Denote by α the index of bargaining power of workers, and we assume that wages are determined according to the maximization of the following Nash products. Generically, this implies:

$$w^j(d) = \text{Argmax}(W_E^j - W_U^j)^\alpha (J^j)^{1-\alpha}$$

where $j = r, f$ reflects the different values of jobs depending on the housing sector. Note that the outside option depends on the housing sector. Agents in the FR sector bargain their wage considering as an outside option the average value of unemployment, where the average is taken from a draw of a location in the area where unemployed workers reside. In the RR sector, they stay where they are, regardless of their employment status. Another important insight is that, in the FR sector, the perfect mobility assumption implies that the asset values of being employed and unemployed are independent of distance, and are from now on denoted by a bar. Bid rents adjust so as to reflect a constant utility value, as is standard in urban economics. Such is not the case in the RR sector, where holding a housing unit at a distance d entails a gain relative to the FR sector and that gain is not independent of distance.

This leads to the following wage equation for workers in the FR sector; we also introduce the notation $r\hat{W}_U^f(d) = (r + \delta_0)\bar{W}_U^f - \delta_0\mathbb{E}_{sec}W_U(d, \bar{d}_u)$ as a deviation from the outside option of workers in the FR sector from its expectation across sectors. The dependence to $\bar{d}_u$ arises from the fact that workers losing their job would relocate to a place at an average distance $\bar{d}_u$:

$$w^f(d) = (1 - \alpha)\{\tau_e d + R^f(d) + r\hat{W}_U^f(d)\} + \alpha[y - aR^f(0)] \quad (13)$$

Now introduce the notation $r\hat{W}_U^r(d) = (r + \delta_0)W_U^r(d) - \delta_0\mathbb{E}_{sec}W_U(d, \bar{d}_u)$ as a deviation of the outside option of workers in the FR sector from its expectation across sectors. Workers in the RR sector get a wage:

$$w^r(d) = (1 - \alpha)\{\tau_e d + R^r(d) + r\hat{W}_U^r(d)\} + \alpha[y - aR^r(0)] \quad (14)$$

where the workers in d have an outside option that preserves their current distance to the city center. The complete calculations in [Online Appendix B.5](#).

Reusing the notation $\underline{\tau}_U$ and $\underline{\tau}_E$ defined earlier on, one can calculate the (negative of) the slopes of bid rents in the FR sector, as:

$$\begin{aligned} -\frac{\partial \Psi_U(d)}{\partial d} &= \tau_u + \frac{\delta_0 p^r}{r + \delta_0 p^f} \left[\underline{\tau}_U - \omega_f \frac{\partial w^r(d)}{\partial d} + \frac{\partial R^r(d)}{\partial d} \right] \\ -\frac{\partial \Psi_E(d)}{\partial d} &= -\frac{\partial w^f(d)}{\partial d} + \tau_e + \frac{\delta_0 p^r}{r + \delta_0 p^f} \left[\underline{\tau}_E - (1 - \omega_s) \frac{\partial w^r(d)}{\partial d} + \frac{\partial R^r(d)}{\partial d} \right] \end{aligned}$$

Proposition 1 on the slope of rents in the city was with fixed wages. These slopes are potentially affected by wage gradients, which themselves depend on the bargaining power of workers. In particular, rent gradients are steeper when workers have a larger bargaining power. We can now complement Proposition 1 as follows:

Proposition 2: complementarity between labor and housing market frictions, with endogenous wages. *Under endogenous wages, the third derivative of bid rents with respect to distance, commuting costs and bargaining power of workers is positive, that is, the (negative of) the slope of rents in the flexible rents sector can be amplified by the wage effects when workers' bargaining power is large; or attenuated when it is low; any flattening in the rent gradient of $R'(d)$:*

1. *raises rents in the FR sector for sufficiently high bargaining power of workers;*
2. *reduces rents in the FR sector for sufficiently low bargaining power of workers.*

In addition, as before, the rent gradients are monotonically decreasing with commuting costs τ_e and τ_u .

The intuition is relatively straightforward. Wages depend on rents, while rents, determined from bid rents, depend on wage gradients as well. The equilibrium embeds these interconnections, and a matrix of a linear system must be inverted (see [Online Appendix B.6](#)). Further, the wage effect on rents is multiplicative in $1 - \alpha$ as can be guessed from the wage equations and can be verified formally in [Online Appendix B.6](#). Therefore, the strength of the wage effect is larger when α is smaller and vice versa.

4.6. Equilibrium

The system is linear in distance to the CBD: this is true for wage functions, utility functions. It is piecewise linear equilibrium rent function in the FR sector. These functions are thus fully characterized by two points over their interval of linearity, e.g. in $d = 1$ and $d = 1 - u$, or $d = 0$ and $d = 1 - u$. Two value functions are only characterized by one point if they are constant, W_U^f and W_E^f . This leads us to 12 endogenous variables and the associated 12 equations. The endogenous variables are, using $d^{sub} = 1 - u$:

$$u, \theta, \psi_u^f(d^{sub}), \psi_e^f(d^{sub}), \psi_e^f(0), \psi_u^f(1), \\ \bar{W}_E^f, \bar{W}_U^f, W_E^r(0), W_E^r(1), W_U^r(0), W_U^r(1).$$

The equations are as follows. A first equation defining θ comes from the labor demand in [Equation \(11\)](#) using that the value of the firm is proportional to the value of the surplus of the worker due to wage bargaining. Two additional equations come from the bid rents for the employed and unemployed in $d = d^{sub} = 1 - u$ from [Equations \(6\) and \(7\)](#); their equality gives a fourth equation. A fifth equation is the equality of rents to agricultural rent in $d = 1$ in the same bid rent equations. Two more equations are provided by the definitions of asset values in the FR sector in [Equations \(2\) and \(3\)](#)—they are identical in space so they can be expressed in any point, in particular in d^{sub} (in these definitions, we substitute the two wage rules (13) and (14)). Four more equations come from the definitions of asset values in the RR sector in [Equations \(4\) and \(5\)](#) expressed in two points ($0, d^{sub}$ or $d^{sub}, 1$). A 12th equation is the steady-state value of unemployment in [Equation \(9\)](#).

The existence and uniqueness of the equilibrium can be derived as follows. The steady-state condition of unemployment defines a negative relation between θ and u ,

$u = s/(s + f(\theta))$, decreasing from 1 to 0 when $\theta \rightarrow \infty$. Reciprocally, we need to find under which condition the entry equation of firms defines a function $\theta^{LD}(u)$ that is upward-sloping and admits positive values. The labor demand (LD) curve is a partial equilibrium relation with 11 variables endogenously solved, using u as a parameter. Since all the 11 variables (value functions, rent and wages entering the right-hand side of (LD)) can be shown to linearly depend on θ and u , it will be easy to characterize the slope of $\theta^{LD}(u)$ with the theorem of implicit functions.

Figure B.1 in [Online Appendix B.7](#) represents both the Beveridge curve (or a version of it, the downward-sloping solid convex curve in θ, u , both defined from [Equation \(9\)](#)) and the labor demand curve in the urban equilibrium.

On uniqueness: In the standard, spaceless equilibrium where θ is determined from labor demand independently of the level of unemployment, it turns out that equilibrium is unique when it exists. In the urban equilibrium, one can prove that θ derived from labor demand is monotonic in u . From simulations, it is increasing with our preferred parameters values (see later in the article, in Section 4.8). This case is represented by the intersection of the upward-sloping curve and the Beveridge curve, itself defined as the long-term negative relation between vacancies and unemployment or sometimes as a link between θ and unemployment. As standard in the search and matching literature, this link arises from the steady-state equality between inflows into unemployment and outflows from that pool of unemployed workers, obtained from [Equation \(9\)](#).

On existence: We only study the benchmark case where the slope of labor demand $\theta^{LM}(u)$ is constant or positive. In that case, the existence arises from a simple viability condition: one simply needs that productivity net of rents and transportation costs of the employed is large enough relative to the utility of unemployment. See [Online Appendix B.7](#) for the analytic expression for the (LD) curve, a proof of its monotonicity and a viability condition.

4.7. A summary of the equilibrium: labor demand, an average wage equation and a rent equation

It is not easy to visualize the equilibrium given the large number of variables and the number of equations involved. However, the nature of the equilibrium can be summarized in three summary variables: the labor market tightness θ that implies a unique unemployment rate; the expected wages, denoted by $\mathbb{E}w$ in the city; and the maximum rent in the center of the city, denoted by $R^{max} = R^f(0)$. As we now show, they are determined as the intersection of two curves in the plane $(\theta, \mathbb{E}w)$ and $(\mathbb{E}w, R^{max})$.

Equilibrium θ is indeed a function of expected profits and therefore expected wages, which is a downward-sloping curve of the expected wage, negatively indexed on the rent paid by firms that lower labor demand, both in the vacancy stage (left-hand side) or in the stage where the firm has a worker producing (right-hand side; recall that $q(\theta)$ is decreasing in θ). This features the labor demand curve, as:

$$(LD) \quad \frac{\gamma + aR^{max}}{q(\theta)} = \frac{y - aR^{max} - \mathbb{E}w}{r + s} \quad (15)$$

On the other side, the expected wage can be shown to be independent of the equilibrium rents paid by workers, but depends on the rent paid by firms. It also positively depends on labor market tightness. After a few steps, one obtains that the expected wage is the sum of the traditional terms in labor search models, augmented by the cost of rents

for firms. There are in addition a few terms arising from the bargaining position of workers with respect to their commuting costs and rents, summarized by the wage-setting curve:

$$\begin{aligned}
 \text{(WS)} \quad \mathbb{E}w = & (1 - \alpha)z + \alpha[y - aR^{\max} + \theta\gamma + \theta aR^{\max}] \\
 & + (1 - \alpha)p^f [R^f(\bar{d}_e) - R^f(\bar{d}_u)] \\
 & + (1 - \alpha)[\mathbb{C} + \delta_0 p^r W_U^r(\bar{d})]
 \end{aligned} \tag{16}$$

with $\mathbb{C} = p^f \tau_e \bar{d}_e + p^r \tau_e \bar{d} - p^f \tau_u \bar{d}_u - p^r \tau_u \bar{d}$. The second line is a term reflecting the differential rents in the FR sector between employed and unemployed workers. The last line is the sum of a term reflecting the differential commuting costs between employed and unemployed workers, and a residual term reflecting the possible transitions in the FR sector that vanish when the share p^r or the transition rate δ_0 disappear. Note, in the first line, that the positive term in $R^{\max}$ comes from the weaker bargaining position of firms if they lost their worker and the negative term comes from their lower profits due to higher rent paid. Hence, for $\theta < 1$, higher rents reduce wage pressures and for $\theta > 1$, they raise them.

Finally, using the slopes of rents calculated in Propositions 1 and 2, one has the rent curve:

$$\begin{aligned}
 \text{(RC)} \quad R^{\max} = & R_A + u \left(\tau_u + \frac{\delta_0 p^r}{r + \delta_0 p^f} \left[\tau_U - \omega_f \frac{\partial w^r(d)}{\partial d} + \frac{\partial R^r(d)}{\partial d} \right] \right) \\
 & + (1 - u) \left(-\frac{\partial w^f(d)}{\partial d} + \tau_e + \frac{\delta_0 p^r}{r + \delta_0 p^f} \left[\tau_E - (1 - \omega_s) \frac{\partial w^r(d)}{\partial d} + \frac{\partial R^r(d)}{\partial d} \right] \right)
 \end{aligned} \tag{17}$$

This formula is complex but since we determined that the model was linear in distance, we have a nice result coming out of it; if wages in the city shift due to a rise in unemployment insurance, the slopes of wages with respect to distance remain constant and the rent varies only marginally, due to changes in employment. The effect is of the order of magnitude of the effect of wages on employment times the slope difference in bid rents of employed and unemployed workers in the FR sector. Hence, the rent curve as a function of expected wages shifted by z is almost vertical in space ($\mathbb{E}w, R^{\max}$).

4.8. Simulation of the three-variable equilibrium

We now explore quantitatively the equilibrium properties of the model with a focus of a natural comparative static exercise in addition: a policy reducing transportation costs reduces the spatial misallocation costs studied so far, in a way that depends on the level of frictions in both labor and housing markets. Figure 3 depicts the relevant curves of the model using the calibration described in Section B.8 in Online Appendix and further explores some comparative exercises. In the left column, we display the labor market equilibrium. The darker (blue with circles) lines are the wage curve as a function of θ at a given rent structure, and the lighter (orange) curves are the equilibrium value of θ for a given wage and rent structure. In the right column, we represent the (wage-rent) equilibrium. The rent curve (darker, blue line) is inelastic while the lighter (orange with triangles) is the wage function given a shift in rents (coming from a shift in agricultural rent that

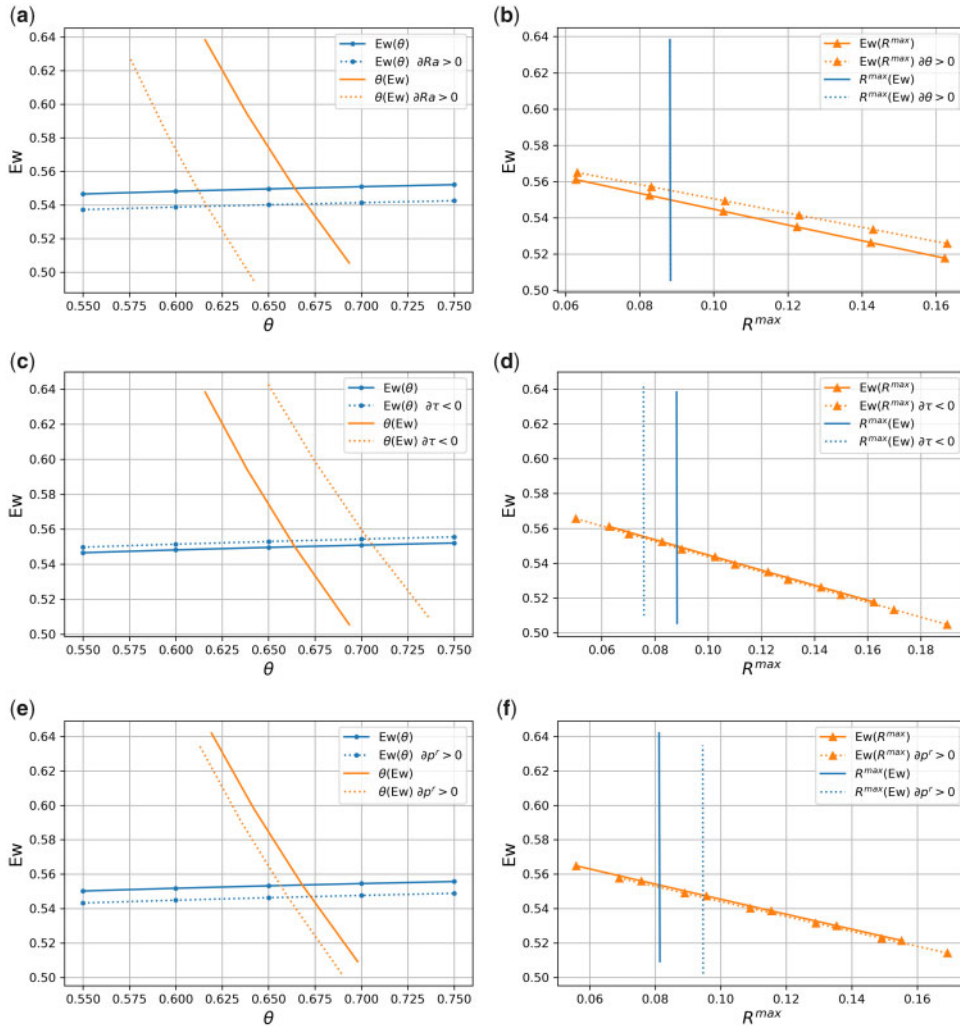


Figure 3. The equilibrium with exogenous population. Panel (a) $Ew - \theta$ increasing R^{max} , Panel (b) $Ew - R^{max}$ increasing θ , Panel (c) $Ew - \theta$ decreasing τ by 20%, Panel (d) $Ew - R^{max}$ decreasing τ by 20%, Panel (e) $Ew - \theta$ increasing p^r by 50 p.p., Panel (f) $Ew - R^{max}$ increasing p^r by 50 p.p.

Source for (a, c, e): Authors' computation. Ew is built shifting θ exogenously while deactivating the labor demand equation. θ is built with an exogenous shift of wages allowing all equations to adjust.

Source for (b, d, f): Author's computation. Ew is built shifting the exogenous agricultural rent. R^{max} is built with an exogenous shift of wages. All equations are allowed to adjust when building both curves.

homothetically shift the rent structure), fixing θ to preserve the partial equilibrium nature of these 2×2 graphs.

In Figure 3(a), we also represent the shift of the curves when the agricultural rents R_a varies (dashed lines) which reduces labor demand and has both a positive and negative effect on wages: the negative effect dominates, but not enough to avoid a decrease in labor

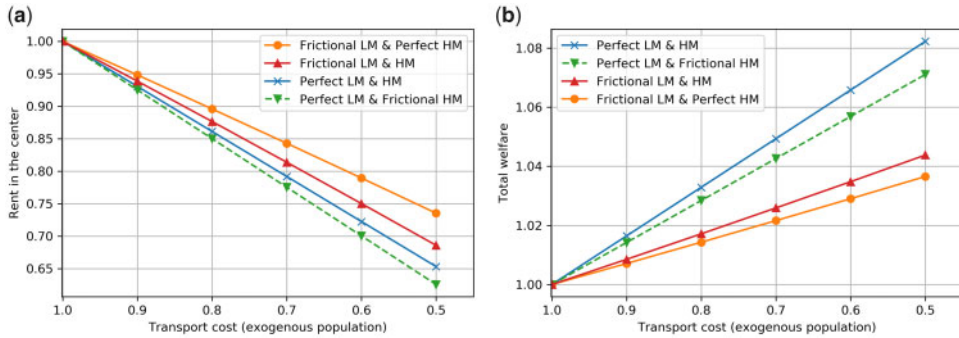


Figure 4. Frictions and transport costs. (a) Flexible rent in the center, (b) welfare.
Source: Author's computation.

market tightness and ultimately in employment. In Figure 3(b), we look instead at a shift in θ on the rent–wage equilibrium. Rents are, as discussed above, invariant in the shift in θ and the wage profile rotates clockwise, leading to higher wages. In addition, the rent curve is a function of the level of unemployment (Figure B.3 in Online Appendix), corresponding to the rent curve in Equation (17).

In Figure 3(c and d), we look at the effect of a transportation policy shock (improvement of 20% of the commuting cost). This raises labor demand with a marginal effect on wages as the term $\mathbb{C}$ is a differential term between employed and unemployed. The same marginal shifts in the wage–rent curve can be seen in the right panel of the second line. As expected, rents decrease and absorb the reduction in commuting costs, as in Propositions 1 and 2. In Figures 3(e and f), we display the effect of a rise in the share of the RR sector, thus removing some efficiency in the housing market. This reduces wage pressures as workers in that sector pay lower rents, but reduces labor demand because it pushes rents in the center and thus raises the production costs for firms. Overall, employment decreases but with a different set of parameters the effect on employment could be easily reversed. The right part of the third line shows that indeed most of the effect of p^r is to raise the rents in the city for those in the FR sector and for firms. This effect has an ambiguous sign. One must therefore study it with simulations. These can be found in Chapelle et al. (2019).

4.9. The complementarity between frictions and a transport policy

We next discuss the role of frictions in both markets on the responsiveness of the equilibrium effect of transportation costs on rents, in showing the improvement of transport costs (reducing homothetically τ_e and τ_u by an index going from 1 to 0.5).

In Figure 4, we report the impact of a decline of τ from its initial benchmark value indexed by 1 to 0.50, in four different cases: the (red upward) triangle curve corresponds to the case of high housing market frictions with $p^r = 0.6$ and high labor market frictions where the scale parameter of the matching function is low, so that the rate of frictional unemployment is maximum.

In contrast, the (blue) crossed line represents the case of lowest frictions in both markets, with $p^r = 0$, so the entire housing market is perfect and a very high scale parameter in matching, so that unemployment is marginally close to zero, featuring a perfect labor market, still with no housing market frictions. The (green dashed downward) triangle line

is the polar case of low labor market frictions and high housing market frictions while the (orange) circle line represents the case with no housing market frictions and high labor market frictions.

Figure 4(a) shows that the maximum impact on rents in the center (the maximum rent) in response to lower transportation costs is maximum (lowest curve) with no labor market frictions, because there are more employed commuting more, and the highest level of housing market frictions, because the misallocation cost of regulated rents is maximal and therefore the improvement of transportation has greater effect on the demand for land. This illustrates Equation (7), Propositions 1 and 2.

Figure 4(b) shows welfare gains. The situation is different here: the improvement in welfare arising from better transportation is partly captured by pre-existing frictions. Therefore, the frictionless case (crossed blue line) is the curve that provides the highest increase in welfare: no frictions dissipate the impact of gains in efficiency of transport. The existence of housing market frictions reduces the gains (green, dashed, downward triangle curve) at $\tau = 0.5$, from approximately 8–7%, in our calibration. However, the largest dissipation is due to labor market frictions: the gains in transportation distort the labor market as increased labor market congestion is raising the cost of opening vacancies.

5. Conclusion

We built a model of a dual housing market and frictions, and study its equilibrium properties. It is designed to show some feedback of housing rents regulations: they push up rents in the non-regulated sector and reduce labor demand; there is also some feedback on wages and overall, the equilibrium is summarized by three main variables: the index of labor demand, the average (expected) wage for firms and the maximum rent in the city (as well as its rent structure). We derive the comparative statics of the model either with exact solutions or with a simulation based on realistic parameters from Paris urban unit.

In particular, we discuss the impact of transportation policies. They reduce transportation costs and lower rents in the center. Transportation improvements also limit the adverse misallocation consequences of both rent controls and social housing. These policies are conflated in our model, but they can be studied separately. An expansion of the social sector raises average commuting costs, which can be compensated by a corresponding reduction in generalized transportation costs. We also find that a larger regulated sector tends to decrease wages and to increase the rent gradient in the flexible rent sector.

Future works should consider several interesting issues left aside as our article is only as a first step to include a dual housing market into the urban analysis. The first issue is segregation by income and employment status. Social housing in our model allows for the local coexistence of different status, but the potential benefits for society, to be weighted against the additional commuting costs at the agglomeration level, are beyond our scope. A second and related issue is that of income heterogeneity; our article could be enriched with different skill categories but again would address a different question than ours; our focus here is only on aggregate employment effects and welfare for a semi-representative agent model where agents can be of four different statuses (employed vs. non-employed and in the private or social housing). As already discussed, the allocation to social housing in France is partly disconnected from income given lax eligibility criteria and sometimes discretionary and political decisions. We do not discuss intercity

commuting and the speed of trains between cities, a dimension that would be interesting to integrate in subsequent works. We also leave aside suburbanization of jobs initially for reasons of tractability but also as a first-order approximation given the local context. In the USA, one can refer to [Baum-Snow \(2007\)](#) for estimates of a sizeable effect of new highways on the size of the local population. We do not discuss that rent controls act as an insurance against pricing monopsony for ongoing leases, generally highly relevant issue in the Anglo-Saxon world. In France, arbitrary rent increases are *de facto* limited by a cap (named *indice de référence des loyers*) generally close to inflation. Hence, in this context, the extension of the size of social housing does not provide any further insurance so that we neglect this issue.¹⁸

Another interesting issue is that of the optimal location of the social sector and more generally of the optimal location of rent controls in general. This issue could be discussed in a more general model where p^r would vary with location. Preliminary calculations in this direction indicate that at some additional complexity costs, one could derive interesting results: it is likely that, absent segregation externalities, the social housing sector should be rather located at the periphery.

Finally, although one could rightly argue that the setup is specific to Paris, we discussed in the text how the benchmark structure can be adapted to other contexts. Further introducing amenities along the lines of [Brueckner \(1990\)](#), or different social housing allocation rules or alternative rent controls would be a fruitful extension of this model.

Supplementary material

[Supplementary data](#) for this paper are available at *Journal of Economic Geography* online.

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18 See [Arnott \(1995\)](#) for a broader perspective on this issue.

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Appendix

A. Extensions

A.1. Implications of other shapes of transport functions

The transport cost function is linear in distance in our framework. In their paper, [Guglielminetti et al. \(2016\)](#) provide empirical evidences based on an exhaustive panel of Austrian unemployed workers and discuss the issue of non-linearity. They found that a good description is a convex then concave function, but that this arises mostly from lowest wage workers. For highest wage workers (above median), the effect is mostly linear. To the extent that the large agglomeration that we represent in the model has rather skilled workers as compared to less rich regions or less dense cities, we consider that linearity is a good starting point. In our present model, a non-linearity in costs would directly translate into a non-linearity in bid rent functions (see e.g. Equations (6) and (7)). In principle, such an extension would not change the features of the equilibrium to the extent that the bid rents cross only once. They may, however, cross twice. Such an effect would be present if for some reasons, the cost of transportation was, for one group, very high close to the center, and very flat far

away from the center (i.e. highly concave), while linear for the other. In that case, in the FR sector, there would be a possibility of having the ‘concave cost’ group out-bidding the ‘linear cost’ group in the center and in the far periphery.¹⁹

A.2. Endogenous consumption of space

There are two possible endogenous choices of space quantities. One is the decision of firms. The second dimension of space choices concerns workers household consumption. We developed the latter extension in a previous version of this text for the purpose of a different quantitative exercise and can provide the model upon request. As regards to the decisions of firms, one could indeed assume that parameter a is the outcome of a choice, where the trade-off is about raising productivity $y(a)$ or reducing turnover $s(a)$ with $y' > 0, y'' < 0$ and $s' < 0, s'' > 0$: better working environment, office space being one of them, may improve workers’ conditions. The cost is to pay higher office space. A simple condition for this choice can be derived, assuming that it can be done only when the position is filled and do not affect future choices of space that will be optimal again in the future. It uses the labor demand curve adapted to the above assumptions and simplifying with exogenous wages w :

$$[(r + s(a) + \delta_0)J^j(d) = y - aR^f(0) - w + \delta_0\mathbb{E}_{sec}J(d)$$

The first-order condition states that at the optimal space choice,

$$R^f(0) = -s'(a)J^j(d) + y'(a)$$

Firms balance out the cost of land to the benefits of giving more space to workers in reducing turnover. Interestingly, the choice, if done *ex post*, depends on the value of the job for the firm: it can be the case that firms give more office space to workers they indeed prefer, as is well known in academia.

A.3. Transport costs externalities

Our benchmark analysis assumes that generalized transport costs are also the social costs. However, there are various congestion effects that can easily be incorporated and their welfare effect would, if anything, only amplify the positive effects of transportation policies and the adverse effects of social housing. Indeed, assume that the generalized individual transportation cost $c(d)$ of the households is a function of her/his individual commuting distance d , but also of the average commuting costs in the agglomeration, denoted by $\bar{C}$ with say, $c(d) = C^\epsilon \tau_K d$ where $\epsilon > 0$ is a congestion effect, $K = u, e$ is the employment status of the person and

$$\bar{C} = p^f[(1-u)\bar{d}_e\tau_e + u\bar{d}_u\tau_u] + \frac{1}{2}(1-p^f)[(1-u)\tau_e + u\tau_u]$$

19 [Zenou \(2009\)](#) explored such a possibility in the case of search unemployment depending on distance. In that case, unemployed workers living close to the center would obtain jobs fast, while those in the periphery would remain unemployed for a longer time.

It is immediately apparent that improving the efficiency of transportation reduces individual costs both directly through τ_K , $K = u, e$ and through the externality effect as long as ϵ is positive, generating a double dividend.

In the analysis in Behrens et al. (2014) and Behrens and Robert-Nicoud (2015), the congestion externality parameter needs not to be large to exert significant effects on the equilibrium. In our endogenous population variant, the cost of commute is linked, along their lines, to the population size, as a larger population leads to higher commuting distances if it also leads to urban sprawl. This therefore leads to a new and interesting discussion on the preference of a social planner for raising urban density (keeping the average distance constant and densifying both the transportation network and the population within the city limits), rather than contributing to urban sprawl.

A.4. Equilibrium with endogenous population and urban sprawl

If we now let L be an endogenous variable, workers relocate to the periphery and contribute to urban sprawl. Other variables adjust. Keep the notation d for a distance index between 0 and 1, 1 still being the periphery, so that all previous equations hold, except that all commuting costs must be multiplied by $L \neq 1$. This corresponds to a long-run expansion of the urban area toward the agricultural land (urban sprawl). Total employment is $E = (1 - u)L$.

In the spirit of the rent determination process, we make the long-run assumption is that the *ex ante* value of unemployment in the urban area is determined by the outside option:

$$\bar{w}_U^f \equiv \bar{V} \quad (\text{A.1})$$

where $\bar{V}$ is a reservation level determined outside the agglomeration. This 13th equation solves for the 13th endogenous variable. The logic of existence and uniqueness is not very different, as the endogenous value of L does not affect the Beveridge relation, hence the logic of Figure B.1 in Online Appendix B.7 is the same, where the (LD) curve is a partial equilibrium relation.²⁰

The extension is useful to fully assess the employment effect of a policy affecting transportation costs as illustrated in Figure B.5 in Online Appendix. Most of the impact indeed comes from the arrival of new households into the city due to the decline in transportation costs and, in the spirit of the previous subsection, of the decline of congestion costs. At constant population, the employment effect of a decline in transportation cost is low, and with endogenous population, the employment effect is larger, and yet, the level of unemployment is not affected, since the new household entry decision is precisely an arbitrage between the values of unemployment in and out of the city. The only possible effect on unemployment could come from an additional labor demand effect not present here: if firms reduced their operating costs or were more productive or if their workers were more effective, then and only then could the rate of unemployment decline in the open city case.

20 The unemployment rate in the city does not need to be equal to the unemployment rate outside the city: this is only the value of unemployment that is equal. The two values depend not only on unemployment but also on structural parameters that may differ across cities.